AHMED CHERIF

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Education

- **Preparatory Institute for Engineering Studies (ranked 61 among 1200 candidates),** IPEIS — *2 years* — *2018-2020*
- **National School of Computer Science ENSI**, Computer Science *engineering diploma* **(ranked 2)** — *3 years* — *2021-2024*
- **Baccalaureate Diploma in mathematical Science**

Experiences:

- **Data Scientist | Sofrecom Tunisia** — **Sfax, Tunisia**

16/10/2024 – Now

IA4archi

I developed an advanced Agent-to-Agent (A2A) communication platform using the Model Context Protocol (MCP) and Google Cloud Platform, enabling distributed AI agents to orchestrate complex workflows and collaborate seamlessly. The system integrates serverless architecture, enterprise-grade security, and comprehensive observability for production-grade agent interactions.

► Architected and deployed a serverless MCP server on Google Cloud Run with auto-scaling capabilities, implementing three core tools for A2A communication, web content fetching, and cloud storage operations.

► Developed a custom Python decorator for automatic BigQuery logging of all MCP tool executions, capturing detailed metrics including arguments, outputs, errors, and execution time, enabling comprehensive observability and analytics across distributed agent systems.

► Designed and delivered a standardized MCP blueprint architecture for Sofrecom Group, establishing enterprise-wide guidelines for agent deployment, security patterns, and cloud-native best practices across the organization. Integrated Open Policy Agent (OPA) for context-aware policy enforcement and developed a decorator-based BigQuery logging system capturing detailed execution metrics, achieving comprehensive observability across all tool operations. Successfully deployed the end-to-end solution into production, thoroughly documenting all processes and preparing for future enhancements in model stability and performance optimization.

► Built and deployed production-ready Vertex AI Agent Engine integrations with HTTP wrapper functions, enabling natural language multi-tool orchestration and seamless agent chaining capabilities.

► Successfully delivered end-to-end solution with extensive test coverage (9+ test suites), deployment automation scripts, and comprehensive documentation, preparing the system for production deployment with auto-scaling and zero-downtime

Keywords: Model Context Protocol (MCP), Agent-to-Agent Communication (A2A), Google Cloud Platform (GCP), Vertex AI Agent Engine, Cloud Run, Open Policy Agent (OPA), BigQuery, Firestore, FastMCP, Python, Docker, Serverless Architecture, REST API, SSE, Transport, Cloud Storage, Policy Enforcement

Smart Fibre Achat

I developed an advanced forecasting system to optimize fiber co-financing over 12 months, enabling Orange to improve profitability in inter-operator agreements. The system integrates complex FTTH park data and uses time series models implemented in Dataiku.

► Engineered comprehensive dataset preparation, including advanced data cleaning, temporal feature creation, and optimized SQL joins to ensure high data quality and processing efficiency.

► Designed and implemented a robust data pipeline architecture, migrating workflows to Dataiku and integrating data flows between OWD, Stargate, and GCP with efficient storage in BigQuery.

► Developed and fine-tuned time series forecasting models using MLForecast, establishing a multi-model prediction workflow and achieving a mean absolute error (MAE) of approximately 6%.

► Transformed and optimized Python scripts into native SQL queries, parallelizing processing tasks and enhancing memory management to significantly improve system performance.

► Successfully deployed the end-to-end solution into production, thoroughly documenting all processes and preparing for future enhancements in model stability and performance optimization.

Keywords: Time Series Forecasting, Dataiku, GCP, BigQuery, SQL, MLForecast, Feature Engineering, Data Pipeline, Pyspark, Python

Parc Énergie et Emplacement

I designed and deployed robust forecasting pipelines for both site-level infrastructure and energy capacity, enabling Orange to anticipate and plan for the migration from copper to fiber technologies and optimize energy management across the national network.

► Developed multi-horizon forecasting pipelines that generate monthly (24 months) and annual (5 years) predictions for install base and energy metrics by geographical location, leveraging advanced data modeling techniques and cloud-native infrastructure (BigQuery, Dataiku, GCP Storage).

► Engineered intricate data preprocessing workflows including extraction from multiple BigQuery sources, comprehensive cleaning, temporal alignment, and aggregation of key metrics by site and period, ensuring data consistency and reliability.

► Consolidated datasets and extended time axes for forecasting, combining technical, commercial, and billing bases with production and cancellation/order flows, while facilitating adjustment of trends and seasonality per technology.

► Established automated visualization outputs and dashboards, providing business stakeholders with intuitive trend analysis by site and technology, and ensuring storage of all predictions and retrainable models in BigQuery and Google Cloud Storage.

► Documented, deployed, and maintained modular codebases for parallel pipelines (energy equipped/ordered, location-based annual/monthly), achieving strong error performance (low MAE) and enabling future scalability, automation, and industrialization of predictive analytics.

Keywords: Time Series Forecasting, BigQuery, Google Cloud Platform (GCP), Dataiku, Cloud Storage, Data Pipeline, Feature Engineering, Multi-horizon Prediction, Python, SQL, Model Deployment, Cloud Architecture

POC – Agentic System for Claims Processing with Generative AI (Ongoing Proof of Concept)

Piloted a Proof of Concept for automating and analyzing infrastructure claims using an agent-based system combined with Google BigQuery ML (BQML) and generative AI techniques.

► Designed an agent-based pipeline to automate and analyze infrastructure claims using multi-source data integrated into BigQuery.

► Developed robust data preprocessing to extract key features from claim files, technician notes, and order details.

► Applied BQML and generative AI for automated pattern detection and anomaly analysis in customer contestations.

► Delivered consolidated outputs with AI-driven insights for each disputed PTO, improving claim validation accuracy.

Documented pipeline logic and architecture to ensure maintainability and scalability.

Keywords: ADK, Generative AI, Claims Analysis, Data Integration, BigQuery, NLP, Data Pipeline, Agentic IA

05/2/2024 - 03/08/2024

I have developed an advanced real-time lip-reading system that translates visual speech cues into audible speech. This system processes live video feeds to generate precise text predictions, which are then synthesized into speech, enhancing communication for individuals.

► Developed a novel real-time lip-reading system using CTC, TimeDistributed, 3DCNN and Bi-LSTM networks.

► Achieved high accuracy with a Character Error Rate (CER) of 8.15% and Character Accuracy (CA) of 91.85% on the GRID dataset.

► Implemented a deployment strategy using Flask to enable real-time lip reading through a web application.

► Achieved text predictions and synthesized speech by normalizing image data, using CTC decoding, correcting text with TextBlob.

► Integrated TensorFlow and MediaPipe to capture and process lip movements.

► Constructed the DATAV1 dataset with more than 600 files for initial model training and evaluation.

► Conducted a comprehensive analysis of deep learning model architectures for lip reading using the DATAV1 dataset.

► Explored and evaluated multiple architectures like ResBlock3D, Conv3D, Conv2D, TimeDistributed, attention mechanism, and LSTM.

► Identified the optimal architecture achieving a peak validation accuracy of 98.18%.

► Contributed insights into effective model selection for accurate lip-reading performance.

► Constructed the DataV2 dataset with more than 4000 files for enhanced model training and evaluation.

► Designed a new lip-reading model architecture incorporating 3D CNN, Bi-LSTM, TimeDistributed, and SoftMax layers.

► Developed a real-time lip-reading application with a video interface displaying synchronized subtitles and predictions.

► Implemented a chatbot using Voice flow and trained it with GPT-3.5 for user interaction and assistance

► Designed and implemented a Video Metting Page

► converting text to speech with Google Text-to-Speech (gTTS).

Keywords: Attention Mechanisms, TimeDistributed, BatchNormalization, MediaPipe, CTC Loss, CTC decoding, Bi-LSTM, 3DCNN, LSTM, 2DCNN, ResBlock3D, Image Normalisation, Wordninja, TextBlob, Google Text-to-Speech (gTTS), Model Selection, TensorFlow, Keras, Flask, Python, SSE, JavaScript, Threading, Generative AI, GPT-3.5, Deep Learning, Computer Vision, Quantum Machine Learning, Spiral Life Cycle

- **Data Scientist | Fysali SAS Bio-incubateur Eurasanté France**

19/7/2023 - 20/09/2023

I have developed an application to prevent medical violence and managed a comprehensive database for a cutting-edge Medical Text Classification Application, leveraging advanced Natural Language Processing (NLP) techniques. The application automatically categorizes medical texts into "non-medical violence" and "Medical violence" categories, providing rapid insights. It serves as a valuable tool for enhancing healthcare communication and awareness.

► Developing an application to prevent medical violence

► Successfully tested and implemented various NLP models to optimize the accuracy.

► Designed and maintained a well-organized database crucial for model training and evaluation.

► Collaborated in the creation of a user-friendly interface, ensuring accessibility to a wide audience.

► Contributed to the mission of improving healthcare transparency and understanding.

Keywords: NLP, Transformers, Transfer Learning, Bert, Distilbert, Universal Encoder, Python, CNN, RNN, LSTM, Python, Streamlit, Data Mining, Deep learning

- **Data Scientist | Mosofty** **Tunis, Tunisia**

01/07/2023 - 30/08/2023

During my summer internship, I worked on a project aimed at enhancing the performance of the company's Optical Character Recognition (OCR) system. My primary role involved developing an artificial intelligence module to improve text recognition accuracy in scanned documents. Additionally, I created an Excel-based database to train and evaluate the model, resulting in a significant reduction in recognition errors. Also, I developed a Flask-based microservice to automatically extract information from CVs. Designed and developed an artificial intelligence module using Flask to enhance the OCR system's performance.

► Utilized the XGBoost algorithm and XMLRegressor for optimizing text recognition accuracy.

► Gathered and prepared relevant data from scanned documents for model training.

► Created and managed an Excel-based database for model training and evaluation.

► Conducted rigorous testing to assess the module's performance and accuracy.

► Extracted structured data from CV documents in various formats.

Keywords: Flask, Docker, Machine Learning, XGBoost, XMLRegressor, Python, Microservice

Research Projects:

- **Google Cloud Data Pipeline Analytics Project:**
  A Python Flask web portal for uploading sales data files stored in a GCS bucket. Automated Cloud Functions handle ETL processes, loading data into BigQuery, with Looker Studio for reporting and dashboards.
  Keywords: Google Cloud Storage, Cloud Functions, BigQuery, Python, Looker, Flask

- **Cricket Statistics Pipeline with Google Cloud Services:**
  I have walked through the intricate steps of constructing a comprehensive cricket statistics pipeline using Google Cloud services. From retrieving data via the API to crafting a dynamic Looker Studio dashboard, each phase contributes to the seamless flow of data for analysis and visualization.
  Keywords: Google Cloud Storage, Cloud Function, Cloud Composer, Python, Dataflow, Apache Airflow, BigQuery, Looker

- **agentic AI systems:**
  As Freelancer I have developed a lot of agentic AI systems, including intelligent agents capable of autonomous decision-making and advanced problem solving across real-world business scenarios.
  Keywords: Agentic AI, Autonomous Agents, LangChain, Langgraph, Generative AI, Python, Flask

- **Ask multiple pdfs using Longchain:**
  The MultiPDF Chat App is a Python application that allows you to chat with multiple PDF documents. You can ask questions about the PDFs using natural language, and the application will provide relevant responses based on the content of the documents. This app utilizes a language model to generate accurate answers to your queries.
  Keywords: Langchain, Generative IA, Streamlit, Python, Embeddings, PyPDF2, FAISS

- **Disease Detector App:**
  The project involves developing an application using AI to analyze images and identify signs of diseases with high precision. One of the key advantages of this approach is that it can be implemented at a low cost, making it accessible to a larger number of people.
  Keywords: Vgg16, Flask, Python, Computer Vision, Deep Learning

Skills Summary

- **Languages** Python, R, SQL
- **Frameworks And Tools** PyTorch, TensorFlow, Keras, NumPy, Pandas, NLTK, OpenCV, Seaborn, Matplotlib, Streamlit, Flask, FastAPI, Uvicorn, Starlette, Docker, Kubernetes, GCP, BigQuery, Dataflow, Cloud Composer, Vertex AI, Vertex AI Agent Engine, BQML, Looker, Cloud Functions, Apache Airflow, Spark, Hadoop, Power BI, Dataiku, NoSQL, Firestore, LangGraph, LangChain, ADK, Cloud Run, Cloud Storage, FastMCP, MCP Protocol, Open Policy Agent (OPA), Cloud Build, SSE (Server-Sent Events), Pyspark, Python
- **Skills** Deep Learning, Machine Learning, Generative AI, Large Language Models (LLMs), Agentic Systems, NLP, Computer Vision, Time Series Forecasting, Feature Engineering, Data Pipeline Design, ETL Processing, MLOps, Model Deployment, Classification, Segmentation, Prediction, Data Integration, Cloud Architecture, SCRUM
- **Soft Skills** Leadership, Project Management, Writing, Team Management, Communication, Coaching

Certification

- **Azure Fundamentals — Certiport/Microsoft** (Issued: Feb 17, 2025) [verify.certiport.com: uhpc-XMSn]
- **Azure Data Fundamentals — Certiport/Microsoft (**Issued: Feb 17, 2025) [verify.certiport.com: wCWSd-48Dm]
- **Azure AI Fundamentals — Certiport/Microsoft** (Issued: Feb 21, 2025) [verify.certiport.com: mUuK-uTCL]
- **Associate Data Engineer** — DataCamp (Certified: Dec 25, 2024)
- **Associate Data Scientist** — DataCamp (Certified: Jan 18, 2025)
- **Data Literacy Fundamentals** — DataCamp (Certified: Dec 25, 2024)
- **Microsoft Azure Fundamentals (AZ-900)** — DataCamp (Completed: Feb 2, 2025)
- **Power Platform Fundamentals — Certiport/Microsoft** (Issued: Mar 5, 2025) [verify.certiport.com: emd7-4wNT]
- **Security, Compliance, and Identity Fundamentals — Certiport/Microsoft** (Issued: Feb 25, 2025) [verify.certiport.com: wLEmQ-Fa8K]
- **365 Fundamentals — Certiport/Microsoft** (Issued: Feb 24, 2025) [verify.certiport.com: wAKHw-2Fq3]
- **Dynamics 365 Fundamentals Finance and Operations Apps (ERP) — Certiport/Microsoft** (Issued: Mar 27, 2025) [verify.certiport.com: mnST-uTEW]
- **Dynamics 365 Fundamentals Customer Engagement Apps (CRM) — Certiport/Microsoft** (Issued: Mar 27, 2025) [verify.certiport.com: mnhW-uTEG]
- **Project Management 101: PMP Certification Training** — (Certified: Feb 21, 2025; Certificate code: 7941375)
- **Fundamentals of Red Hat Enterprise Linux** — Coursera/Red Hat
- **AI For Everyone** — Coursera/DeepLearning.AI
- **Object-Oriented Data Structures in C++** — Coursera/University of Illinois at Urbana-Champaign